Bird Classification System using Audio Signal Based on Deep Learning

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Abstract— **Bird species identification plays a crucial role in ecological maintenance and bird biodiversity conservation. Traditional methods rely on visual observation and sound analysis by field experts, which is really time-consuming and requires extensive knowledge. This study presents a bird species classification system using audio in the form of sounds (call, alarm, song) using various machine learning and deep learning approaches, integrated into a web-based application to facilitate practical use in real life. This experiment uses six different modeling approaches and to be tested and compared, which are: Random Forest, Support Vector Machine (SVM), XGBoost, Feedforward Neural Network (FNN), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU). The dataset consists of audio recordings of 20 unique bird species sourced from publicly available repositories, preprocessed and divided into 80% training data and 20% testing data, and Mel-Frequency Cepstral Coefficients (MFCC) are used as the main feature extraction method. Experimental results show that XGBoost achieves the highest classification accuracy (79%), followed by Random Forest (78%) and SVM (77%), performed slightly better than deep learning models in this dataset setting. These findings show that the ensemble decision tree method is highly effective for MFCC-based bird audio classification, while the deep learning model shows promise with larger datasets and further hyperparameter tuning. The best-performing model is being integrated into a web application to support researchers and conservationists in real life situations.**

Keywords— **bird classification, audio signal processing, MFCC, XGBoost, deep learning, web-based application**

I. Introduction

Identifying species of birds in the wild has quite an important part of maintaining diversity in the nature, ecological, and conservation parts. Traditionally, identifying birds has been done visually with direct bird watching and analyzing birds sounds by the professionals. Sadly these methods require quite a lot of time and in-depth knowledge to know and recognize the difference between species. With the growing technology of Artificial Intelligence and machine learning, bird noise classification systems automatically start expanding and improving to help the identification process much faster and efficiently than before [1].

However, bird sound classification still remains a challenge due to various factors such as audio noise in audio recordings process, differential variations in song patterns between individual birds, and similarities in song characteristics between species [2]. The recording quality and environmental conditions during recording can also affect the performance of classification models. Therefore, selecting the right extraction feature from audio to processable model method and machine learning model is crucial in the making of an accurate bird song classification system.[3].

In several previous studies, various machine learning and deep learning approaches have been used to classify bird calls. Deep learning models such as Convolutional Neural Networks (CNNs) are often used because they can automatically extract complex features from audio signals [4]. In audio processing, features such as Mel-Frequency Cepstral Coefficients (MFCC) and Mel Spectrograms are widely used to represent the frequency characteristics of bird calls, enabling them to be processed by machine learning models [5], [6]. Spectrogram-based approaches also enable the use of image-based classification techniques to analyze bird call patterns more effectively [7].

Based on this, this study develops an artificial intelligence-based bird call classification system using several different modeling approaches. The four models used in this study are XGBoost, Multi-Layer Perceptron (MLP), Convolutional Neural Network (CNN) with MFCC features, and CNN with Mel Spectrogram representation. This study aims to compare the performance of these four models in classifying bird calls and determine the most effective model for use in an AI-based bird call detection system.

II. Literature Review

Birds play a vital role in maintaining ecosystem balance, acting as seed dispersers, pest controllers, and indicators of environmental health. Changes in bird populations can be used as early indicators of environmental degradation and climate change. Research by Baowaly et al. [8] shows that the presence and distribution of bird species are closely related to the ecological conditions of an area, making bird monitoring a crucial aspect of biodiversity conservation. Therefore, efficient methods are needed to automatically identify bird species on a large scale.

One widely used approach is audio-based bird classification, as each bird species has unique call patterns. Dwarkadas et al. [9] explain that the use of audio signals in classification allows the system to perform identification without the need for direct visual observation, making it more effective in complex environmental conditions. This process generally involves feature extraction from the audio signal, such as Mel-Frequency Cepstral Coefficients (MFCC) and spectrogram-based representations.

MFCC is one of the most commonly used feature extraction methods in audio classification. Research by Janah et al. [10] showed that MFCC is able to represent the frequency characteristics of bird sounds effectively, making it suitable for use as input for machine learning models. This is reinforced by research by Pattraksha et al. [11] which shows that the combination of MFCC with classification algorithms such as K-Nearest Neighbor (KNN) is able to produce quite good performance in recognizing bird species based on sound.

In addition to manual feature-based methods, various machine learning algorithms have also been used in bird classification. A study in the journal Applied Acoustics [12] compared several methods, such as Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), and Convolutional Neural Network (CNN), and showed that each method has its own advantages depending on the type of data used. MLP, as one of the basic neural network methods, has been shown to perform quite well in spectrogram-based sound classification, as demonstrated in another study in the same journal [13]. However, MLP performance still depends on the quality of the provided features.

Developments in deep learning have brought significant improvements in audio classification, particularly with the use of Convolutional Neural Networks (CNN). CNNs are able to automatically extract features from input data without requiring manual feature engineering. Recent research [14] shows that the use of CNNs in bird sound classification yields higher performance than traditional methods. This is supported by another study in the journal Ecological Indicators [15], which shows that the use of a CNN architecture with Mel Spectrogram input can significantly improve classification accuracy.

Although various methods have been developed, several challenges remain in audio-based bird classification. These include vocal variation within a species, environmental noise, and dataset limitations. Furthermore, direct comparisons between machine learning and deep learning methods within a single study are limited. Therefore, this study aims to compare the performance of several classification models, such as XGBoost, MLP, and CNN, in classifying bird calls and evaluate the best method for application-based system implementation.

III. Methodology

A. Data Gathering and Pre-processing

The dataset was sourced from the "Sound of 114 Species of Birds" repository via KaggleHub. The data consists of audio recordings in MP3 format along with metadata containing common names, scientific names, and geographic locations. In this study, we used 20 recordings of the main bird classes in the dataset to ensure statistical significance and facilitate classification. The audio was recorded at a rate of 22,050 Hz, so we removed the parts at the start and end of the recordings using the `librosa.effects.trim` function with a threshold of 23 dB so we could focus on the actual bird sounds.

B. Feature Extraction

To transform audio waves into representations suitable for machine learning, several audio signal processing techniques were used, including:

- *Short-Time Fourier Transform (STFT):* Used to convert time-domain signals into the frequency domain with a frame size of 2048 and a hop length of 512.
- *Mel-Frequency Cepstral Coefficients (MFCCs):* The main feature used is 40 MFCCs per frame to capture the sound power spectrum based on human ear perception (Mel scale). For the classical model, the average MFCC is calculated over time, while for the sequential model, the order of the time series is maintained.
- *Other Spectral Features:* Spectral Centroid and Zero-Crossing Rate are analyzed to understand the brightness and noise level characteristics of bird calls

C. Classification Models

Six different architectures were selected, implemented and compared with data from the top 20 bird classes:

- *Random Forest (RF):* An ensemble of decision trees used as a baseline classifier, operating on mean MFCC features.
- *Support Vector Machine (SVM):* Uses a Radial Basis Function (RBF) kernel with standardized features to find the optimal hyperplane for species separation.
- *XGBoost:* A Gradient Boosting framework implemented to improve accuracy through sequential decision tree construction.
- *Feedforward Neural Network (FNN):* A multilayer perceptron with a dropout layer (0.3) to prevent overfitting of the extracted statistical features.
- *LSTM:* A recurrent architecture with two layers of 64 units designed to capture temporal dependencies in audio sequences, followed by a dense classification layer.
- *GRU:* A gated recurrent architecture with a structure similar to LSTM, also using two layers of 64 units, offering a computationally lighter alternative for sequence modeling.

D. Experiment Setup

Initial experiments were conducted in the Google Colab program environment using several libraries and exporting datasets from Kaggle into Colab. The data was split into two parts: one for training and one, for testing. We used a way to make sure both parts had the same amount of each type of data. The training part was eighty percent (80%) of the information and the testing part was twenty percent (20%).

We used a computer program to teach the deep learning model for thirty rounds. This program used something called the Adam optimizer and the Sparse Categorical Cross-entropy loss function. To see how well the deep learning model was doing we looked at the accuracy, precision, recall and F1-score metrics of the deep learning model.

E. Chart Diagram

To see the clear chart diagram, visit this link

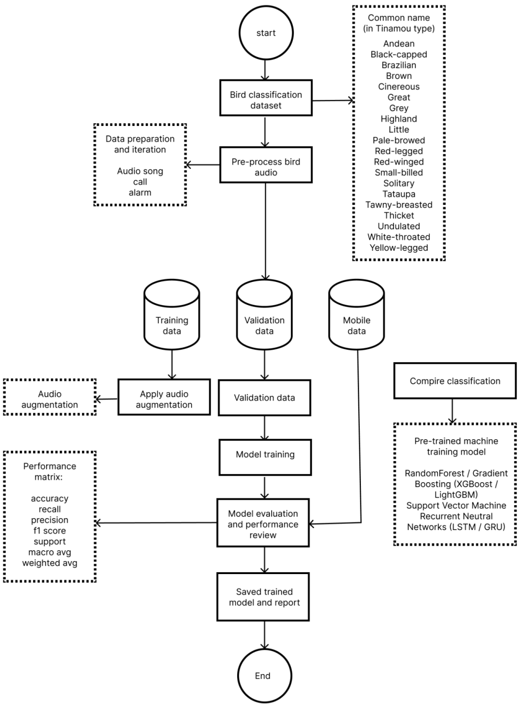

Fig. 3.5 Chart Diagram Image

IV. Result and Evaluation

A. Model Performance Comparison

The following table summarizes the overall performance of all six models across accuracy and macro-averaged metrics:

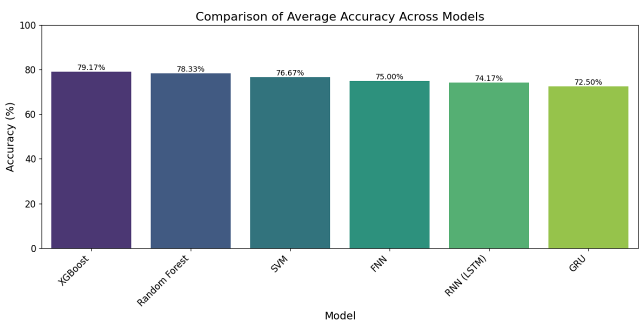

Fig. 4.1 Overall Model Performance Summary

Based on the test results on the dataset, the classification reports (precision, recall, F1-score) for each model is as follows:

1) *XGBoost*

--- LAPORAN KLASIFIKASI: XGBoost ---

	precision	recall	f1-score	support
Andean Tinamou	0.67	1.00	0.80	6
Black-capped Tinamou	1.00	0.67	0.80	6
Brazilian Tinamou	0.62	0.83	0.71	6
Brown Tinamou	0.80	0.67	0.73	6
Cinereous Tinamou	1.00	1.00	1.00	6
Great Tinamou	0.50	0.50	0.50	6
Grey Tinamou	1.00	0.83	0.91	6
Highland Tinamou	0.71	0.83	0.77	6
Little Tinamou	0.75	1.00	0.86	6
Pale-browed Tinamou	0.60	0.50	0.55	6
Red-legged Tinamou	0.80	0.67	0.73	6
Red-winged Tinamou	1.00	0.83	0.91	6
Small-billed Tinamou	0.83	0.83	0.83	6
Solitary Tinamou	0.43	0.50	0.46	6
Tataupa Tinamou	1.00	1.00	1.00	6
Tawny-breasted Tinamou	1.00	1.00	1.00	6
Thicket Tinamou	0.83	0.83	0.83	6
Undulated Tinamou	1.00	1.00	1.00	6
White-throated Tinamou	0.75	0.50	0.60	6
Yellow-legged Tinamou	0.83	0.83	0.83	6
accuracy			0.79	120
macro avg	0.81	0.79	0.79	120
weighted avg	0.81	0.79	0.79	120

Fig 4.1.1 Classification report of XGBoost Model

XGBoost achieved the highest overall accuracy of 79%, with macro-averaged precision, recall, and F1-score all at 0.79–0.81. Several species such as Cinereous Tinamou, Tataupa Tinamou, Tawny-breasted Tinamou, and Undulated Tinamou were classified with a perfect F1-score of 1.00, reflecting clear and distinctive acoustic features. Conversely, species such as Great Tinamou (F1: 0.50) and Solitary Tinamou (F1: 0.46) showed lower performance, likely due to high intraspecific vocal variation or feature overlap with other species.

2) *Random Forest*

--- LAPORAN KLASIFIKASI: Random Forest / Gradient Boosting ---

	precision	recall	f1-score	support
Andean Tinamou	0.62	0.83	0.71	6
Black-capped Tinamou	0.75	0.50	0.60	6
Brazilian Tinamou	0.83	0.83	0.83	6
Brown Tinamou	0.67	1.00	0.80	6
Cinereous Tinamou	0.75	1.00	0.86	6
Great Tinamou	1.00	0.83	0.91	6
Grey Tinamou	0.83	0.83	0.83	6
Highland Tinamou	0.83	0.83	0.83	6
Little Tinamou	1.00	1.00	1.00	6
Pale-browed Tinamou	1.00	0.50	0.67	6
Red-legged Tinamou	0.67	0.67	0.67	6
Red-winged Tinamou	0.71	0.83	0.77	6
Small-billed Tinamou	0.83	0.83	0.83	6
Solitary Tinamou	0.60	0.50	0.55	6
Tataupa Tinamou	0.71	0.83	0.77	6
Tawny-breasted Tinamou	0.83	0.83	0.83	6
Thicket Tinamou	0.71	0.83	0.77	6
Undulated Tinamou	1.00	1.00	1.00	6
White-throated Tinamou	1.00	0.33	0.50	6
Yellow-legged Tinamou	0.71	0.83	0.77	6
accuracy			0.78	120
macro avg	0.80	0.78	0.78	120
weighted avg	0.80	0.78	0.78	120

Fig 4.1.2 Classification report of Random Forest Model

Random Forest achieved 78% accuracy, performing comparably to XGBoost. Species such as Little Tinamou and Undulated Tinamou were classified with perfect F1-scores of 1.00, while White-throated Tinamou and Solitary Tinamou yielded lower F1-scores of 0.50 and 0.55, respectively, indicating ambiguity in their feature distributions.

3) Support Vector Machine (SVM)

--- LAPORAN KLASIFIKASI: SVM ---

	precision	recall	f1-score	support
Andean Tinamou	0.71	0.83	0.77	6
Black-capped Tinamou	1.00	0.50	0.67	6
Brazilian Tinamou	0.83	0.83	0.83	6
Brown Tinamou	1.00	0.67	0.80	6
Cinereous Tinamou	1.00	1.00	1.00	6
Great Tinamou	0.67	0.67	0.67	6
Grey Tinamou	0.62	0.83	0.71	6
Highland Tinamou	0.56	0.83	0.67	6
Little Tinamou	0.86	1.00	0.92	6
Pale-browed Tinamou	1.00	0.50	0.67	6
Red-legged Tinamou	0.80	0.67	0.73	6
Red-winged Tinamou	0.56	0.83	0.67	6
Small-billed Tinamou	0.62	0.83	0.71	6
Solitary Tinamou	0.57	0.67	0.62	6
Tataupa Tinamou	1.00	0.83	0.91	6
Tawny-breasted Tinamou	1.00	0.67	0.80	6
Thicket Tinamou	0.62	0.83	0.71	6
Undulated Tinamou	1.00	1.00	1.00	6
White-throated Tinamou	0.75	0.50	0.60	6
Yellow-legged Tinamou	1.00	0.83	0.91	6
accuracy			0.77	120
macro avg	0.81	0.77	0.77	120
weighted avg	0.81	0.77	0.77	120

Fig 4.1.3 Classification report of SVM Model

SVM achieved 77% accuracy. Cinereous Tinamou and Undulated Tinamou were perfectly classified (F1: 1.00), suggesting clearly separable features in high-dimensional space. Pale-browed Tinamou and White-throated Tinamou showed lower recall and F1-scores, indicating ambiguity in their MFCC feature representations.

4) Feedforward Neural Network (FNN)

--- LAPORAN KLASIFIKASI: FNN ---

	precision	recall	f1-score	support
Andean Tinamou	0.50	0.67	0.57	6
Black-capped Tinamou	1.00	0.50	0.67	6
Brazilian Tinamou	0.71	0.83	0.77	6
Brown Tinamou	0.67	1.00	0.80	6
Cinereous Tinamou	1.00	1.00	1.00	6
Great Tinamou	0.75	0.50	0.60	6
Grey Tinamou	0.62	0.83	0.71	6
Highland Tinamou	0.62	0.83	0.71	6
Little Tinamou	1.00	1.00	1.00	6
Pale-browed Tinamou	1.00	0.50	0.67	6
Red-legged Tinamou	0.67	0.33	0.44	6
Red-winged Tinamou	0.83	0.83	0.83	6
Small-billed Tinamou	0.80	0.67	0.73	6
Solitary Tinamou	0.50	0.33	0.40	6
Tataupa Tinamou	1.00	0.83	0.91	6
Tawny-breasted Tinamou	0.71	0.83	0.77	6
Thicket Tinamou	0.62	0.83	0.71	6
Undulated Tinamou	1.00	1.00	1.00	6
White-throated Tinamou	1.00	0.67	0.80	6
Yellow-legged Tinamou	0.60	1.00	0.75	6
accuracy			0.75	120
macro avg	0.78	0.75	0.74	120
weighted avg	0.78	0.75	0.74	120

Fig 4.1.4 Classification report of FNN Model

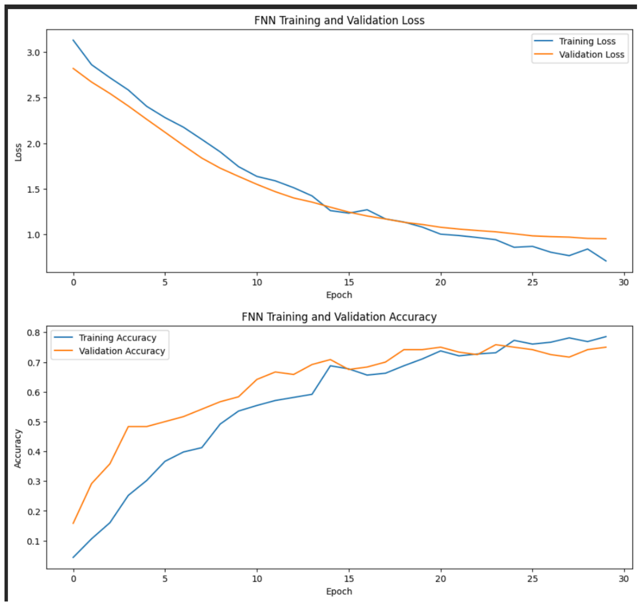

Fig 4.1.5 Training validation loss and accuracy on FNN

FNN achieved 75% accuracy. While species such as Little Tinamou, Cinereous Tinamou, and Undulated Tinamou were well-classified (F1: 1.00), others struggled — Solitary Tinamou (F1: 0.40) and Red-legged Tinamou (F1: 0.44) in particular. The training loss curve showed steady convergence, while the validation loss began fluctuating after several epochs, indicating mild overfitting. Validation accuracy plateaued after initial improvement, suggesting limited generalization to unseen data.

5) Recurrent Neural Network (LSTM)

--- LAPORAN KLASIFIKASI: RNN (LSTM) ---

	precision	recall	f1-score	support
Andean Tinamou	0.75	1.00	0.86	6
Black-capped Tinamou	0.83	0.83	0.83	6
Brazilian Tinamou	1.00	0.67	0.80	6
Brown Tinamou	0.86	1.00	0.92	6
Cinereous Tinamou	0.67	1.00	0.80	6
Great Tinamou	0.75	0.50	0.60	6
Grey Tinamou	0.71	0.83	0.77	6
Highland Tinamou	0.56	0.83	0.67	6
Little Tinamou	1.00	0.83	0.91	6
Pale-browed Tinamou	0.50	0.33	0.40	6
Red-legged Tinamou	0.67	0.67	0.67	6
Red-winged Tinamou	0.60	1.00	0.75	6
Small-billed Tinamou	0.67	0.67	0.67	6
Solitary Tinamou	1.00	0.50	0.67	6
Tataupa Tinamou	1.00	0.83	0.91	6
Tawny-breasted Tinamou	0.60	0.50	0.55	6
Thicket Tinamou	0.67	0.67	0.67	6
Undulated Tinamou	0.86	1.00	0.92	6
White-throated Tinamou	0.60	0.50	0.55	6
Yellow-legged Tinamou	1.00	0.67	0.80	6
accuracy			0.74	120
macro avg	0.76	0.74	0.73	120
weighted avg	0.76	0.74	0.73	120

Fig 4.1.6 Classification report of RNN Model

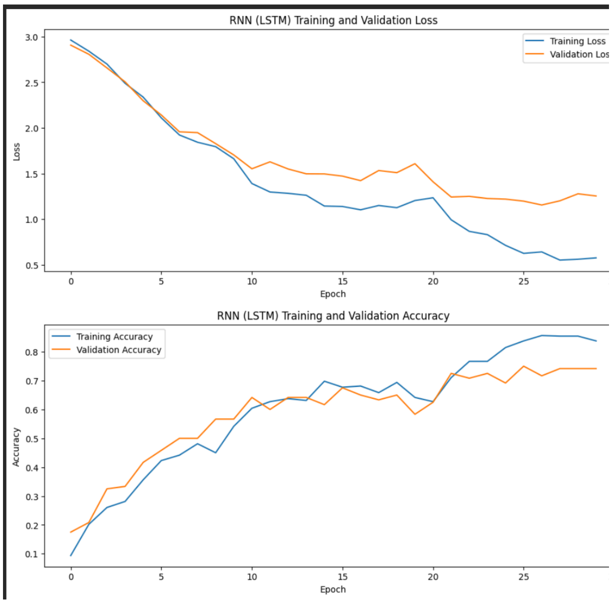

Fig 4.1.7 Training validation loss and accuracy on LSTM

LSTM achieved 74% accuracy. Despite its capacity for temporal dependency modeling, performance was uneven across species — Brown Tinamou and Undulated Tinamou reached F1 of 0.92, while Pale-browed Tinamou scored only 0.40. The validation loss showed mild overfitting after the initial decrease, and validation accuracy fluctuated after peaking, suggesting the model had limited generalization. The dataset size (600 samples) likely constrained the LSTM's ability to fully leverage temporal patterns.

6) *Gated Recurrent Unit (GRU)*

--- LAPORAN KLASIFIKASI: RNN (LSTM) ---

	precision	recall	f1-score	support
Andean Tinamou	0.75	1.00	0.86	6
Black-capped Tinamou	0.83	0.83	0.83	6
Brazilian Tinamou	1.00	0.67	0.80	6
Brown Tinamou	0.86	1.00	0.92	6
Cinereous Tinamou	0.67	1.00	0.80	6
Great Tinamou	0.75	0.50	0.60	6
Grey Tinamou	0.71	0.83	0.77	6
Highland Tinamou	0.56	0.83	0.67	6
Little Tinamou	1.00	0.83	0.91	6
Pale-browed Tinamou	0.50	0.33	0.40	6
Red-legged Tinamou	0.67	0.67	0.67	6
Red-winged Tinamou	0.60	1.00	0.75	6
Small-billed Tinamou	0.67	0.67	0.67	6
Solitary Tinamou	1.00	0.50	0.67	6
Tataupa Tinamou	1.00	0.83	0.91	6
Tawny-breasted Tinamou	0.60	0.50	0.55	6
Thicket Tinamou	0.67	0.67	0.67	6
Undulated Tinamou	0.86	1.00	0.92	6
White-throated Tinamou	0.60	0.50	0.55	6
Yellow-legged Tinamou	1.00	0.67	0.80	6
accuracy			0.74	120
macro avg	0.76	0.74	0.73	120
weighted avg	0.76	0.74	0.73	120

Fig 4.1.8 Classification report of GRU Model

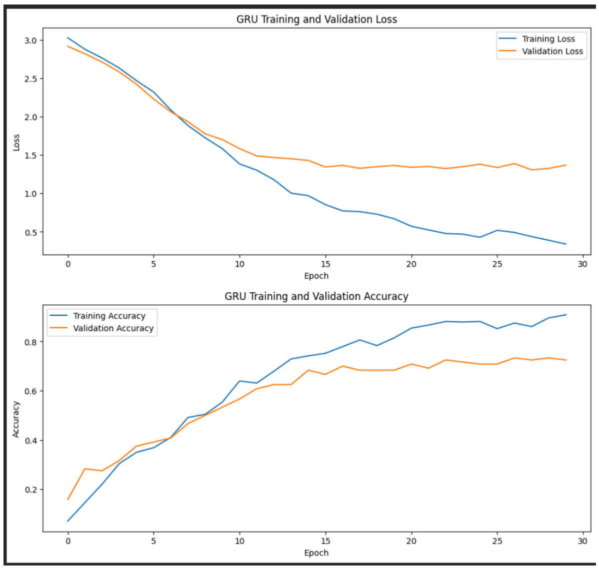

Fig 4.1.9 Training validation loss and accuracy on GRU

GRU achieved the lowest accuracy at 72%, with performance patterns similar to LSTM. Strong results were observed for Cinereous Tinamou and Little Tinamou (F1: 1.00), while White-throated Tinamou scored only 0.44. The training and validation loss curves exhibited similar overfitting patterns to LSTM, with validation accuracy plateauing after early improvement.

B. *Evaluation*

The results clearly show that ensemble tree-based models outperformed deep learning approaches in this experiment. XGBoost and Random Forest achieved the highest accuracy, demonstrating their effectiveness in classifying bird species from MFCC features. Their ability to handle multidimensional feature vectors and generate complex decisions through aggregation of multiple trees proved advantageous for this dataset.

SVM also performed competitively at 77%, confirming that MFCC features — when standardized — are sufficiently separable in high-dimensional space using a nonlinear RBF kernel. Deep learning models (FNN, LSTM, GRU) achieved lower accuracy of 75%, 74%, and 72% respectively. Several factors likely contributed to this result:

- *Dataset size* : While there are only 600 total samples (20 classes × 30 recordings), the dataset is insufficient for deep learning models to fully develop their representational capacity and avoid overfitting.
- *Tuning parameter* : The number of units in recurrent layers, dropout rates, and learning rate schedules were not exhaustively optimized, potentially limiting model performance.
- *Sequence length* : Audio segments were fixed at 5 seconds and padded or truncated to 216 frames, which may not consistently capture all relevant temporal patterns across species.

V. Conclusions

This study put into practice and assessed six different classification models for the purpose of identifying bird species automatically by using audio signals alongside MFCC feature extraction. The ensemble methods XGBoost achieved a performance level of 79 percent and Random Forest reached 78 percent, showing a strong fit for tasks related to audio classification based on MFCC. The SVM also showed competitive results at 77 percent, while the deep learning models which included FNN, LSTM, and GRU performed somewhat less effectively due to limitations related to the size of the dataset and the extent of hyperparameter optimization that was possible.

These results indicate that for datasets that are small to medium in size and that use handcrafted features such as MFCC, methods that are ensemble-based and kernel-based might perform better than architectures based on deep learning. Nonetheless, deep learning models possess significant potential for enhancement when it comes to larger datasets, strategies for data augmentation, and a more thorough search for hyperparameters.

The model that performed the best, which was XGBoost, was integrated into a mobile web-based application that is intended to serve as an accessible tool for researchers who work in the field and for conservationists. Future efforts will aim to increase the dataset to include more species, to incorporate data augmentation techniques, and to investigate end-to-end deep learning architectures like CNN that use Mel Spectrogram inputs in order to achieve better classification performance.

Author Contribution

Richard Paulus Wihardja proposed the idea, finding the datasets, helped writing the paper, and developed the software. **Valent Steve Asman** performed the experiments, collected the data, evaluated the experiment result and the main writer for the paper. **Rialan Dumaris Sibombing** helped writing the paper, designed a poster for this research, and managed the plan & resource. **Abdul Haris Rangkuti** as guidance, supervisor, and advisor for this paper. All authors reviewed and approved the final manuscript.

Data Availability

The raw datasets used and/or analyzed and experimented during this study are available and publicly accessible at https://www.kaggle.com/datasets/soumendraprasad/sound-of-114-species-of-birds-till-2022/data. The 20-class subset used for this experiment is also available at https://www.kaggle.com/datasets/valentasman/sound-of-20-birds-classes

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